

Problem & Data — The AI Backbone

From Problem to Trained Model — Complete Guide

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1. Problem Understanding

The #1 skill in AI is NOT coding or math. It is understanding the PROBLEM.

Golden Rule: Don't start with "which model?" Start with "what would a HUMAN do?"

Turn Vague Goals into Specific Tasks

Vague Goal	Ask What Specifically?	Specific Tasks
Use AI for hospital	What decisions do doctors struggle with?	Predict disease, spot abnormal reports, smart ICU alerts
Make store smarter	What costs you money/time?	Predict demand, detect fraud, group customers
Build ExamGuard	What does cheating look like?	Phone on desk, looking sideways, unusual behavior
AI for cricket	What does the coach DO?	Analyze opponent, predict form, field placement

Watch the Human Expert

For each task, watch/imagine a human doing it. Note: what data they use, how they think.

Define Success BEFORE Building

Question	ExamGuard Example
What does success look like?	Detect 90% of phone usage
Acceptable failure?	Max 5% false alarms
Who is the user?	Invigilator gets alert on screen
Worst case mistake?	Missing real cheating > false alarm

2. The 3 Questions — Core Skill

For EACH task a human does, ask these 3 questions:

Q1: What DATA is the human using? (Images / Text / Numbers)

Q2: HOW is the human thinking? (Recognizing / Sensing abnormal / Grouping / Balancing)

Q3: Does the human have PAST EXAMPLES? (Yes=Supervised / No normal=Anomaly / No tries=RL)

Combination Table

Q1 Data	Q2 Thinking	Q3 Labels?	ML Type	Model Family
IMAGES	Recognizing	YES	Supervised	CNN / YOLO
IMAGES	Something off	NO	Anomaly	Autoencoder
TEXT	Categorizing	YES	Supervised	Naive Bayes / BERT
NUMBERS	Predicting word	YES	Supervised	Classification (LR/DT/RF)
NUMBERS	Predicting number	YES	Supervised	Regression (Linear/Poly/RF)
NUMBERS	Something off	NO	Anomaly	Isolation Forest
NUMBERS	Grouping	NO	Unsupervised	K-Means / DBSCAN
ANY	Balancing tradeoffs	Rewards	RL	Reinforcement Learning

ML vs Simple Rules

If you can write it as one IF-ELSE → don't use ML.

Scenario	ML or Rule?	Why
Temperature > 38 = fever	Simple Rule	One number, one threshold
Predict diabetes from 20 factors	ML	Too many interacting factors
Order > Rs 1 lakh = approval	Simple Rule	One number, one threshold
Detect fraud patterns	ML	Complex pattern across many variables

Practice: Restaurant Manager

Task	Data	Thinking	Labels?	ML Type
Predict chicken to buy	NUMBERS	Predicting amount	YES	Regression
Categorize reviews	TEXT	Good/bad/average	YES	Text Classification
Unusual electricity	NUMBERS	Not normal	NO	Anomaly Detection

bill				
Set lunch prices	NUMBERS+JUDGMENT	Balance profit vs customers	NO	RL
Spot theft on CCTV	IMAGES	Recognize cash in hand	YES	CNN/YOLO

3. Data Collection

4 sources in order: Already Have > Public Dataset > Pre-trained Model > Collect Yourself

Source 1: You Already Have It

Most businesses sit on unused data: hospital records, website databases, CCTV footage, sensor logs.

Source 2: Public Datasets (FREE)

Platform	Best For
Kaggle	Everything — 50K+ datasets
Roboflow	Image/video with labels
HuggingFace	Text/NLP datasets
Google Dataset Search	Search engine for datasets

Source 3: Pre-trained Models (Zero Data!)

Model	Already Does	Data Needed
YOLOv8	Detect 80 objects (phone, person...)	ZERO
MediaPipe Face	Face landmarks, gaze direction	ZERO
MediaPipe Pose	Body skeleton (33 points)	ZERO
BERT	Text understanding	ZERO

Source 4: Collect Yourself

Match your REAL setup: same camera angle, lighting, room. Minimum 200 for transfer learning.

4. Data Validation

5 Checks before using any dataset:

- SIZE: Count rows/images. Enough for your model?
- LOOK: Open 30 random samples. Clear? Labels correct?
- RELEVANCE: Does it match YOUR situation? Street phone != desk phone!
- BALANCE: 500 phone + 500 no phone = good. 4500+50 = bad.
- TRUST: Downloads? Comments? Ratings? Others trained on it?

5. Data Augmentation

Make small data bigger without collecting more:

Technique	What it does	ExamGuard Example
Flip	Mirror image horizontally	Phone on left vs right of desk
Rotate	Tilt 5-30 degrees	Camera slightly tilted
Brightness	Lighter or darker	Morning vs evening exam
Zoom	Closer or further	Different camera distances
Noise	Add grain/blur	Cheap camera simulation
Crop	Different framing	Phone partially visible

500 images x 6 techniques = 3000 images. But real data > augmented data.

6. Data Cleaning

ML engineers spend 80% time on data, 20% on models.

Problem	Fix
Missing values	Delete row or fill with average
Duplicates	Find and remove
Wrong labels	Check random 50, fix mistakes
Outliers (age=500)	Detect and remove or cap
Wrong format ("Rs 50,000")	Convert to numbers (50000)
Inconsistent ("Male","M","m")	Standardize to one format
Blurry images	Remove or recapture
Wrong file types	Convert to standard format

7. Pre-trained Models

Already trained on millions of images. Download and use immediately.

The 10-Photo Test

- Take 10 photos from YOUR actual setup
- Run pre-trained model on them
- Check: >85% correct = use as-is. 60-85% = fine-tune. <60% = wrong model.

Transfer Learning

Pre-trained YOLO + your 300 exam photos = 70% jumps to 93%!

Extending Functionality

YOLO detects phones. Want chits too? Add labeled chit images + fine-tune = now detects both.

8. Data Labeling

Label Type	What it does	When to use
Classification	Whole image = "phone"	Is there a phone? Yes/No
Bounding Box	Rectangle around phone	WHERE is the phone? (YOLO needs this)
Segmentation	Exact outline pixel by pixel	Medical imaging, self-driving
Text Label	"positive" / "negative"	Review sentiment

Tools: Roboflow (web), Labelling (desktop), CVAT (video)

Best practices: Be consistent, label everything visible, have 2 people check.

9. Data Pipeline — Complete Flow

- 1. Problem Understanding
- 2. The 3 Questions
- 3. Data Collection
- 4. Data Validation
- 5. Data Cleaning
- 6. Data Labeling
- 7. Data Augmentation
- 8. Train/Test Split (80/20)
- 9. Model Selection
- 10. Training
- 11. Testing
- 12. If bad → back to step 3. If good → Deploy!

Time spent: 60% data work, 20% training, 10% testing, 10% deployment

10. Multi-Model Systems

Real AI = multiple independent models + regular code connecting them.

ExamGuard Architecture

Module	Model	Data	Output
Phone Detector	YOLO	Camera images	phone/no phone + confidence
Gaze Tracker	MediaPipe	Face video	looking direction + duration
Behavior Monitor	Autoencoder	Full body video	anomaly score 0-100
Identity Verifier	Face CNN	Face vs ID card	match/mismatch
Alert Decision	RL or rules	Scores from 1-4	alert / keep watching

Key Rules

- Different models = different data (each gets only what it needs)
- Each model trained separately (never share training data)
- Connector = regular Python code, NOT ML
- Build ONE module at a time (never all at once)
- Start with rules, add ML later (v0.1 = simple rules, v1.0 = RL)

Build Phases

Phase	What to build
v0.1 (Week 1-2)	Phone Detector only. Simple rule: phone>80% = alert
v0.2 (Week 3-4)	Add Gaze Tracker. Rule: sideways >30s = alert
v0.3 (Week 5-6)	Add Behavior Monitor. Rule: anomaly >70 = flag
v0.4 (Week 7-8)	Add Identity Verifier. Run once at exam start
v1.0 (Week 9-12)	Add RL Alert. Replaces simple rules. Balances all.